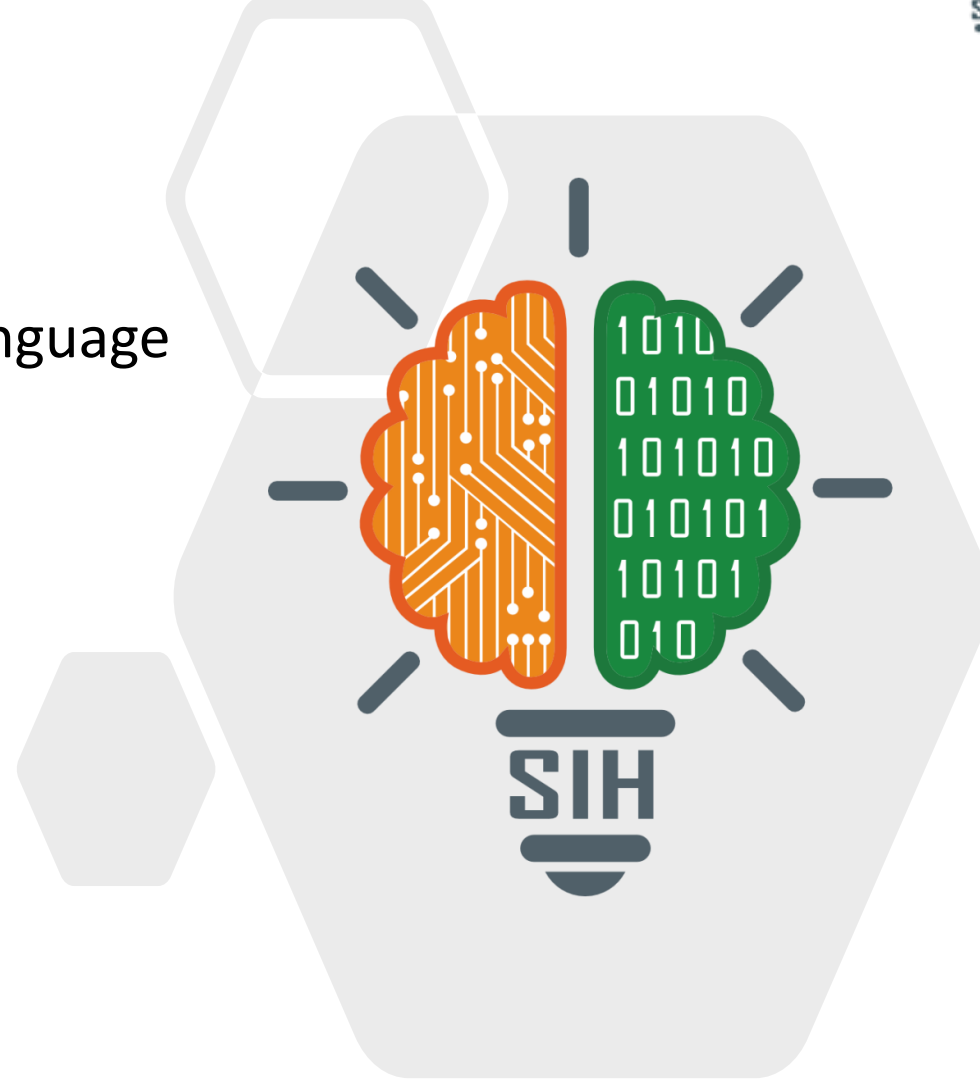


SMART INDIA HACKATHON 2025



- **Problem Statement ID** –SIH25104
- **Problem Statement Title-** Language
Agnostic Chatbot
- **Theme-** Smart Education
- **PS Category-** Software
- **Team ID-** 99
- **Team Name** - LUMO



LANGUAGE AGNOSTIC CHATBOT



- **Detailed explanation of the proposed solution-** A Flask-based web chatbot using the API to handle queries in any language and return well-formatted, code-friendly responses.
- **How it addresses the problem-** *It solves* the problem by providing a single platform that understands and responds to queries in multiple languages without needing separate language specific systems.
- **Innovation and uniqueness of the solution** -A multilingual chatbot with automatic code formatting and responsive design.

- **Technologies used** -Python (Flask), HTML, CSS, JavaScript, API, and a standard web server or cloud hosting platform.
- **Methodology and process for implementation**-User
→ Web UI → Flask Server → API → Flask Server → Web UI → User

FEASIBILITY AND VIABILITY



- **Analysis of the feasibility of the idea**-The idea is highly feasible using readily available web technologies and the API, requiring only standard hosting and minimal infrastructure.
- **Potential challenges and risks**-Possible challenges include API usage costs, rate limits, and ensuring data privacy for multilingual user inputs.
- **Strategies for overcoming these challenges**-Implement request throttling, secure data handling, and optimize API usage with caching and monitoring to control costs and protect privacy.

IMPACT AND BENEFITS



- **Potential impact on the target audience**-It provides users with seamless multilingual assistance and instant code-ready responses, enhancing accessibility and productivity.
- **Benefits of the solution** -It boosts global communication and learning efficiency, reduces development costs, and supports broader digital inclusion.

RESEARCH AND REFERENCES



- **FlaskDocumentation:** <https://flask.palletsprojects.com> – Official guide for building Python web applications.
- **API Docs:** <https://ai.google.dev> – Reference for integrating Google's Generative AI services.
- **MDN Web Docs:** <https://developer.mozilla.org> – Comprehensive resources for HTML, CSS, and JavaScript development.